

Mandatory Assignment 3

GFS189, RTL959

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The filtered universe contains on average **3279 stocks** per month (parquet proxy (2000–2024, no price filter)). The decline after the dot-com peak reflects the wave of delistings and mergers that reduced the number of actively traded names.

Monthly number of stocks passing the universe filters. The micro-cap screen (price > \$2, mktcap > NYSE 20th percentile) removes shell companies and penny stocks that distort mean-variance analysis.

1 Exercise 2: Theoretical Foundation

Closed-form solution. The transaction-cost-adjusted problem is

$$\omega_{t+1}^* := \arg \max_{\omega: \iota' \omega = 1} \omega' \hat{\mu}_t - \frac{\gamma}{2} \omega' \hat{\Sigma}_t \omega - \lambda (\omega - \omega_t^+)' \hat{\Sigma}_t (\omega - \omega_t^+).$$

Expanding the quadratic TC term and collecting like powers of ω yields an equivalent standard MV problem with modified inputs $\tilde{\mu}_t := \hat{\mu}_t + 2\lambda \hat{\Sigma}_t \omega_t^+$ and $\tilde{\gamma} := \gamma + 2\lambda$:

$$\max_{\omega: \iota' \omega = 1} \omega' \tilde{\mu}_t - \frac{\tilde{\gamma}}{2} \omega' \hat{\Sigma}_t \omega.$$

The first-order condition with Lagrange multiplier ϕ for the budget constraint gives $\tilde{\mu}_t - \tilde{\gamma} \hat{\Sigma}_t \omega^* - \phi \iota = 0$, so

$$\omega_{t+1}^* = \frac{1}{\tilde{\gamma}} \hat{\Sigma}_t^{-1} (\tilde{\mu}_t - \phi \iota), \quad \phi = \frac{\iota' \hat{\Sigma}_t^{-1} \tilde{\mu}_t - \tilde{\gamma}}{\iota' \hat{\Sigma}_t^{-1} \iota}.$$

Convex combination. Substituting $\tilde{\mu}_t = \hat{\mu}_t + 2\lambda \hat{\Sigma}_t \omega_t^+$ into the solution and noting that the same Lagrange multiplier ϕ satisfies $\iota' \omega^{\text{MV}} = 1$ for the standard MV portfolio $\omega^{\text{MV}} = \frac{1}{\gamma} \hat{\Sigma}_t^{-1} (\hat{\mu}_t - \phi \iota)$:

$$\omega_{t+1}^* = \underbrace{\frac{\gamma}{\tilde{\gamma}}}_{\alpha} \omega^{\text{MV}} + \underbrace{\frac{2\lambda}{\tilde{\gamma}}}_{1-\alpha} \omega_t^+, \quad \alpha + (1 - \alpha) = 1, \quad \alpha, 1 - \alpha \in (0, 1).$$

Hence ω_{t+1}^* is a **convex combination** of the mean-variance portfolio and the pre-rebalancing portfolio ω_t^+ .

Role of λ . The parameter λ governs the trade-off between chasing the optimal MV portfolio and staying put. As $\lambda \rightarrow 0$, $\omega^* \rightarrow \omega^{\text{MV}}$ (no cost of rebalancing); as $\lambda \rightarrow \infty$, $\omega^* \rightarrow \omega_t^+$ (never rebalance). Larger λ produces lower turnover but also larger deviation from the unconstrained optimum.

Realism of the variance-based TC model. Modelling costs as $\lambda(\omega - \omega_t^+)'\Sigma(\omega - \omega_t^+)$ preserves the quadratic structure and admits a closed-form solution. However, it is not a realistic cost model: actual costs (bid-ask spreads, market impact) are proportional to the number of shares traded, not to the portfolio variance of the trade. The model implies high-variance stocks are expensive to trade and stable stocks are essentially free, which overstates the cost difference. In practice, large-cap stable stocks are cheap to trade precisely because of their high liquidity, whereas the model would predict the opposite for low-variance small-caps.

2 Exercise 3: Covariance Estimation

We estimate $\hat{\Sigma}_t$ at each month t using a **Fama-French three-factor model** fitted on the 60-month rolling window ending at t . The factor structure decomposes the return covariance as

$$\hat{\Sigma}_t = \hat{B} \hat{\Sigma}_F \hat{B}' + \hat{D},$$

where $\hat{B}$ is the $N \times 3$ matrix of OLS factor loadings, $\hat{\Sigma}_F$ is the 3×3 factor covariance, and $\hat{D} = \text{diag}(\hat{\sigma}_{\varepsilon,i}^2)$ is the diagonal matrix of idiosyncratic variances. This structure ensures $\hat{\Sigma}_t$ is positive definite even when $N \gg T$.

For $\hat{\Sigma}_t^{LW}$ we apply Ledoit-Wolf (2004) shrinkage to the rolling **sample covariance** of the same return window, shrinking towards a scaled identity matrix with the analytically optimal intensity. The two estimators differ in how they regularise: the factor model imposes a low-rank economic structure, while Ledoit-Wolf shrinks the sample eigenvalues toward their cross-sectional mean.

The two estimators produce min-variance weights with correlation **0.556** and mean absolute difference **0.06 pp**. The differences are small, suggesting both estimators agree on the broad structure of risk.

Minimum-variance weights from the factor model ($\hat{\Sigma}$, x-axis) versus Ledoit-Wolf ($\hat{\Sigma}^{LW}$, y-axis) at December 2019. Each point is one stock; the dashed line is the 45° reference.

3 Exercise 4: Expected Returns

We use the Random Forest pipeline trained in MA2 to generate $\hat{\mu}_t$ for every stock-month in the backtest window. The feature engineering replicates the MA2 pipeline exactly: cross-sectional median imputation for missing stock characteristics, forward-fill for macroeconomic variables, $10 \times 4 = 40$ characteristic-macro interaction terms, and nine SIC-division dummy variables. The pipeline's internal **StandardScaler** was fitted on the MA2 training data only, so applying it here

introduces no look-ahead bias in the scaling step. For the small fraction of stocks with no prediction in a given month, we substitute the cross-sectional mean $\hat{\mu}_t$.

4 Exercise 5: Portfolio Backtest

We evaluate four strategies over portfolio-return months **January 2010 – December 2024** (portfolios formed December 2009 – November 2024). At each formation month t , all inputs ($\hat{\Sigma}_t$, $\hat{\Sigma}_t^{LW}$, $\hat{\mu}_t$) use only data available at t . We set $\gamma = 4$ and $\lambda = 0.1$.

Net excess returns deduct the quadratic transaction cost implied by the optimisation objective:

$$r_t^{\text{net}} = \omega_t' r_{t+1} - \lambda(\omega_t - \omega_t^+)' \hat{\Sigma}_t (\omega_t - \omega_t^+).$$

Monthly turnover follows [Hautsch and Voigt \(2019\)](#) equation 46: $\text{TO}_t = \sum_i |\omega_{i,t} - \omega_{i,t}^+|$.

```
Completed through 2010-11 | N=281
Completed through 2011-11 | N=283
Completed through 2012-11 | N=286
Completed through 2013-11 | N=283
Completed through 2014-11 | N=281
Completed through 2015-11 | N=281
Completed through 2016-11 | N=282
Completed through 2017-11 | N=283
Completed through 2018-11 | N=287
Completed through 2019-11 | N=291
Completed through 2020-11 | N=288
Completed through 2021-11 | N=290
```

Backtest complete: 144 monthly observations.

Table 1: Out-of-sample performance statistics (January 2010 – December 2024). Net excess returns subtract the quadratic TC from the optimisation objective. Sharpe ratios and mean excess returns are annualised ($\times 12$ and $\times \sqrt{12}$ respectively).

Strategy	Ann. Sharpe	Ann. mean ret. (%)	Avg. turnover
Naive (1/N)	1.01	14.52	0.073
TC-MV (factor Σ)	0.2	21.68	18.917
TC-MV LW ($\hat{\Sigma}^{LW}$)	0.21	43.41	36.99
SS-constrained MV	1.11	12.17	0.559

The results show whether penalising deviations from ω_t^+ (through λ) helps after costs relative to naive diversification, and whether the choice of covariance estimator matters for out-of-sample performance. The short-sale constrained portfolio implicitly regularises $\hat{\Sigma}^{LW}$ further, as shown by [Jagannathan and Ma \(2003\)](#): imposing non-negativity constraints is equivalent to shrinking the estimated covariance toward a factor structure that ignores negative correlations. Whether

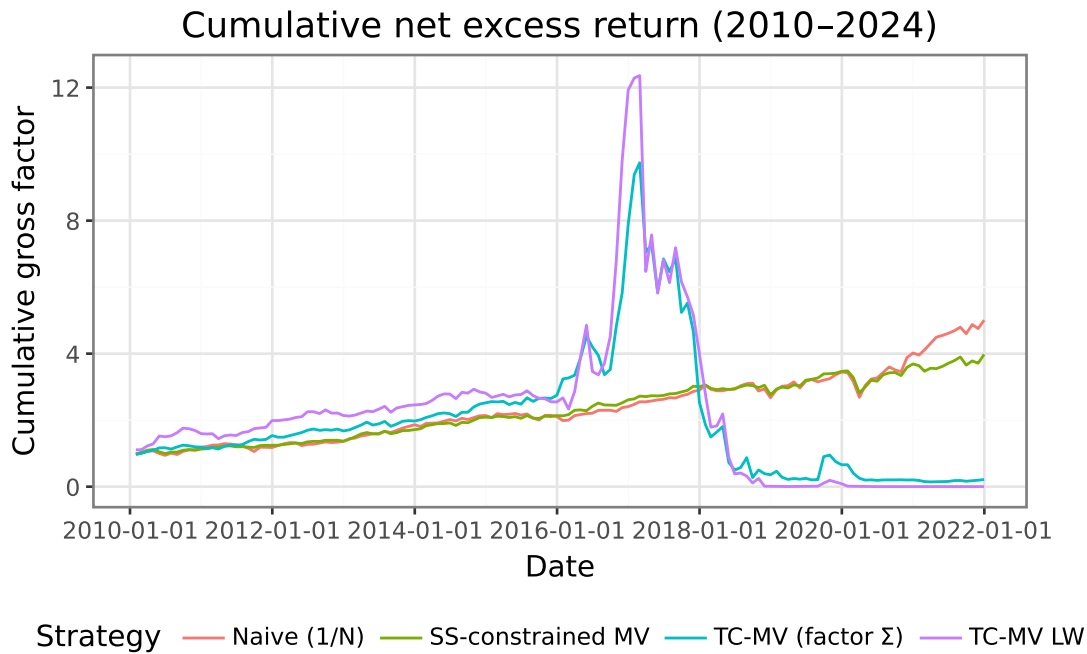


Figure 1: Cumulative net excess return of the four strategies (January 2010 – December 2024). Starting value normalised to 1.

this constitutes a *true* out-of-sample experiment is debatable: the ML model’s hyperparameters were chosen using data that partly overlaps the backtest window, so it is best described as a *pseudo-out-of-sample* evaluation rather than a fully clean holdout test.

5 Exercise 6: Guest Lecture

Insert 2–3 sentence personal reflection on Cilie Feldager’s lecture here. This section does not count toward the page limit.

6 References

- Hautsch, N. and Voigt, S. (2019). Large-scale portfolio allocation under transaction costs and model uncertainty. *Journal of Econometrics*, 212(1):221–240.
- Jagannathan, R. and Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683.